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Team Name: Beyonders			
Team Number: - 39		**College Name: -R V College of Engineering**	
Sl. No.	**Name**	**Contact Number**	**Email ID**
1.	Nihar Mandahas	94828 08895	niharmandahas.cs22@rvce.edu.in
2.	Mohith S	76765 57268	mohiths.cs22@rvce.edu.in
3.	Pranav Darshan	88671 61440	pranavdarshan.cs22@rvce.edu.in
4.	Pratheek Rao MP	86605 01765	pratheekrmp.cs22@rvce.edu.in
5.	Raghuveer Rajesh	83100 54684	raghuveernr.cs22@rvce.edu.in

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Abstract:

In the realm of education, examination plays a very significant role. If writing an exam is considered tedious, think about the process of evaluating hundreds of answer scripts. It usually takes weeks to evaluate all the answer scripts, and there is always a factor of bias in the human corrections. Hence, to eliminate all these problems, we can use a large language model to automate this task and eliminate the human bias. In this project, we have fine-tuned an LLM to evaluate answer scripts related to the subject of operating systems using our handcrafted dataset. Along with the LLM, we use RAG (Retrieval Augmented Generation) to get the context of a given question from a prescribed textbook. The proposed platform also has the capability of analyzing handwriting from an actual answer script, and then this is passed onto the model as input along with context from RAG. Finally, all the features were integrated into an interactive web platform using the AWS SageMaker to deploy the model. By combining all the technology mentioned, we have made a sincere attempt to solve the burden of correcting manuscripts.

1. Introduction:

Examinations are a crucial part of education, but evaluating hundreds of answer scripts manually is time-consuming and prone to human bias. This study proposes an AI-powered system that uses a fine-tuned Large Language Model (LLM) to automate the assessment of answer scripts of any subject. To ensure context-aware evaluation, the model integrates Retrieval-Augmented Generation (RAG) for fetching relevant content from prescribed textbooks. Handwritten scripts are digitized and processed using handwriting recognition. The system is deployed as a web platform via AWS SageMaker.

In addition to text-based answers, the model evaluates technical diagrams using visual understanding techniques. A consolidated student feedback dashboard offers performance insights across all subjects, helping learners identify strengths and weaknesses. This holistic approach reduces manual effort, improves evaluation speed, and ensures consistency. The platform aims to revolutionize the academic assessment process. Overall, it presents a scalable, unbiased solution for modern education.

2. Literature Review

The implementation of automated answer script evaluation systems has gained significant attention due to their potential to reduce grading time and minimize human bias. The use of large language models (LLMs) and retrieval-augmented generation (RAG) frameworks in such systems has been extensively explored in recent studies.

Touvron et al. (2023) introduced Llama2, a fine-tuned LLM designed for open foundation and chat-based applications. The model leverages advanced language understanding capabilities, making it suitable for tasks such as text scoring and contextual analysis through integration with RAG pipelines. The study highlights the model's scalability and efficiency, particularly when deployed using quantization techniques like QLoRA, optimizing memory usage without compromising performance [1].

Alikaniotis et al. (2016) investigated the application of neural networks for automatic text scoring. Their study demonstrated the effectiveness of neural networks in evaluating short-answer responses, emphasizing the importance of training models on domain-specific datasets to enhance scoring accuracy [2].

Olowolayemo et al. (2018) proposed a text similarity measurement approach for short answer scoring, focusing on grammar evaluation in English texts. Their method integrates natural language processing (NLP) techniques to compare student responses with model answers, achieving notable accuracy in grammar-based assessments [3].

Selvi and Banerjee (2010) developed the Automatic Short-Answer Grading System (ASAGS), which assesses student answers by comparing them against predefined reference answers. The system applies text matching and semantic analysis to score responses, demonstrating the potential of automated grading in educational contexts [4].

Kadupitiya et al. (2016) explored the evaluation of short answers in Sinhala language using corpus-based similarity measures. The study underscores the challenges associated with scoring responses in low-resource languages and the importance of incorporating contextual analysis to improve scoring accuracy [5].

Pribadi et al. (2017) proposed a word overlapping method for automatic short-answer scoring, wherein the model identifies key terms and calculates the degree of overlap with reference answers. This method, though effective in factual assessments, is less robust in assessing higher-order thinking skills [6].

Chakraborty et al. (2016) introduced a fuzzy spelling evaluator designed for eLearning systems. The model assesses spelling accuracy using fuzzy logic, making it effective in identifying minor spelling errors and providing corrective feedback [7].

Rababah and Al-Taani (2017) applied automated scoring techniques to Arabic short answers, focusing on improving reliability through natural language processing methods. The study emphasizes the significance of language-specific models in enhancing grading consistency across diverse linguistic contexts [8].

Vij et al. (2020) employed a WordNet-based text similarity approach for automated short-answer evaluation. The model utilizes WordNet graphs to identify semantic similarities between student responses and reference texts, demonstrating high precision in content-based assessments [9].

Lajis et al. (2019) presented the NCI Evaluation framework, designed to assess higher-order thinking skills in short free-text responses. The framework integrates NLP techniques to evaluate reasoning and critical thinking, expanding the scope of automated grading beyond factual assessments [10].

The existing literature underscores the efficacy of integrating LLMs, RAG pipelines, and NLP techniques in automated answer script evaluation. The proposed system leverages the advanced capabilities of Llama2 and RAG to provide context-sensitive grading while maintaining scalability and robustness through AWS SageMaker deployment.

Title	Publication Details	Abstract	Research Gap
Llama2: Open Foundation and Fine-Tuned Chat Models	Touvron, H. et al., arXiv preprint, 2023 [1]	The paper introduces Llama2, a large language model fine-tuned for chat applications. It emphasizes model scalability, quantization using QLoRA, and integration with retrieval-augmented generation (RAG) for context-sensitive responses.	Focused on chat-based applications; limited application to educational grading and context analysis specific to academic datasets.

Automatic Text Scoring Using Neural Networks	Alikaniotis, D., Yannakoudakis, H., Rei, M., arXiv preprint, 2016 [2]	The study applies neural networks to assess short-answer responses, highlighting the importance of domain-specific datasets in enhancing scoring accuracy.	Lacks integration of context retrieval mechanisms, limiting the model's capability to assess content alignment with textbook references.
Short Answer Scoring in English Grammar Using Text Similarity Measurement	Olowolayemo, A., Nawi, S.D., Mantoro, T., IEEE ICCED, 2018 [3]	The proposed method evaluates grammar-based short answers using text similarity metrics, achieving notable accuracy in English grammar assessments.	Limited to grammar evaluation; lacks semantic analysis and contextual feedback generation.
Automatic Short-Answer Grading System (ASAGS)	Selvi, P., Banerjee, A.K., arXiv preprint, 2010 [4]	ASAGS utilizes text matching and semantic analysis to grade short answers based on predefined model responses, aiming for consistency in grading.	Focused on factual answers; lacks adaptability for open-ended responses and higher-order thinking skills assessment.
Sinhala Short Sentence Similarity Measures for Short Answer Evaluation	Kadupitiya, J.C.S., Ranathunga, S., Dias, G., 2016 [5]	The study implements corpus-based similarity techniques to assess short answers in Sinhala, emphasizing linguistic challenges in low-resource languages.	Language-specific; lacks generalizability to diverse academic datasets and multilingual contexts.
Automatic Short Answer Scoring Using Word Overlapping Methods	Pribadi, F.S., Adji, T.B., Permanasari, A.E., AIP Conf. Proc., 2017 [6]	The paper presents a word overlapping technique to assess short answers, focusing on keyword identification	Limited to factual assessments; lacks handling of higher-order thinking and reasoning

		and content matching.	questions.
Intelligent Fuzzy Spelling Evaluator for eLearning Systems	Chakraborty, U.K., Konar, D., Roy, S., Choudhury, S., Education and Information Technologies, 2016 [7]	The model applies fuzzy logic to evaluate spelling accuracy in eLearning systems, providing corrective feedback for minor errors.	Focused solely on spelling evaluation; lacks content analysis and grading capabilities.
Automated Scoring Approach for Arabic Short Answer Essays	Rababah, H., Al-Taani, A.T., IEEE ICIT, 2017 [8]	This study applies NLP techniques to score Arabic short answers, aiming to improve grading consistency across linguistic contexts.	Language-specific; limited generalizability to non-Arabic contexts and diverse subject areas.
Automated Evaluation of Short Answers Using WordNet Graphs	Vij, S., Tayal, D., Jain, A., Wireless Personal Communications, 2020 [9]	A WordNet-based text similarity method is proposed to identify semantic similarities between student responses and reference texts, achieving high precision.	Focused on text similarity; lacks contextual analysis and integration of reference textbooks.
NCI Evaluation: Assessment of Higher Order Thinking Skills via Short Free Text Answer	Lajis, A., Nasir, H., Aziz, N.A., IEEE ICSIMA, 2019 [10]	The framework assesses higher-order thinking skills using NLP techniques to evaluate reasoning and critical thinking in short text responses.	Limited focus on factual assessments; lacks comprehensive grading structure for academic exams.

3. Main objectives:

The main objective of this study is to develop an AI-powered platform that automates the evaluation of handwritten answer scripts, minimizing human bias and significantly reducing the time required for assessment. By leveraging a fine-tuned LLaMA language model and the multimodal LLaMA Vision model, the system aims to accurately interpret both textual and diagrammatic responses. Through the integration of Retrieval-Augmented Generation (RAG), it ensures context-aware grading by referencing prescribed academic content. The platform also consolidates performance insights across subjects into a unified feedback dashboard, providing students and educators with actionable analytics. Ultimately, the objective is to create a scalable, unbiased, and intelligent assessment system tailored for modern educational needs.

A. Develop a fine-tuned LLM model for answer script evaluation.
B. Implement a RAG pipeline to provide contextual analysis.
C. Develop a dashboard for students and teachers to review marks.
D. Deploy the solution using AWS SageMaker for scalability.
E. Enable students to view detailed feedback based on the evaluation, enhancing transparency and learning outcomes.
F. Allow teachers to adjust grading criteria based on textbook content, enabling flexible assessment approaches.

4. Theory and Concepts:

To build a comprehensive automated answer script evaluation system, this study integrates advanced AI theories in natural language processing and computer vision. A fine-tuned large language model (LLaMA) forms the backbone for evaluating textual responses with high contextual accuracy. Unlike traditional systems that use OCR for handwriting recognition, this platform utilizes a multimodal LLaMA Vision model to directly extract and interpret text from scanned answer scripts. Retrieval-Augmented Generation (RAG) enhances this by supplying relevant textbook content to support informed grading. Together with diagram understanding and a consolidated feedback interface, the system provides a scalable, unbiased, and intelligent solution for academic evaluation.

A. Fine-Tuned LLaMA for Answer Evaluation

a. A customized **fine-tuned LLaMA model** is used for grading textual responses.
b. Trained on domain-specific data (e.g., Operating Systems) to understand student answers, compare with expected solutions, and score them accordingly.
c. Reduces subjectivity and increases consistency in evaluation.

B. **Retrieval-Augmented Generation (RAG)**

a. Integrates textbook-based retrieval to provide contextual information related to each question.
b. Enhances the LLaMA model's understanding by grounding answers in prescribed academic materials.
c. Helps ensure fair grading even when students paraphrase or partially recall answers.

C. **Handwriting Recognition using LLaMA Vision**

a. Utilizes the **LLaMA Vision model** to extract and interpret handwritten text directly from scanned answer scripts.
b. Eliminates the need for traditional OCR by leveraging **multimodal capabilities** that combine image understanding with natural language processing.
c. Accurately processes diverse handwriting styles, ensuring clean and structured input for the grading pipeline.
d. Enhances the robustness of the evaluation system, especially for handwritten academic responses.

D. **Diagrammatic Evaluation via LLaMA-4 Scout**

a. Implements **meta-llama/llama-4-scout-17b-16e-instruct**, a multimodal model capable of processing **both text and image inputs**.
b. Used to evaluate diagrams such as memory hierarchies, process scheduling charts, and other visual responses in Operating Systems.
c. Understands diagram intent, labels, structure, and correctness with high accuracy.

E. **Deployment using AWS SageMaker**

a. The entire pipeline is integrated and deployed using **AWS SageMaker**, ensuring scalable compute and secure API endpoints.
b. Enables real-time evaluation and efficient resource management.

F. **Consolidated Student Feedback Dashboard**

a. A centralized platform that visualizes scores and insights across multiple subjects.
b. Provides students with personalized feedback, trends, and suggested improvements.
c. Useful for educators to track class-level performance and knowledge gaps.

5. Project objective and methodology:

The primary objective of this project is to develop an intelligent, AI-powered platform that automates the evaluation of handwritten answer scripts across various academic subjects, thereby reducing the time, effort, and bias inherent in manual grading. To achieve this, the methodology involves collecting and preprocessing handwritten and diagrammatic answer scripts using the multimodal LLaMA Vision model to extract textual and visual content accurately. The core evaluation leverages a fine-tuned LLaMA language model, optimized using QLoRA, to assess answers in context with relevant textbook information retrieved through a Retrieval-Augmented Generation (RAG) approach powered by FAISS and MiniLM4 embeddings. The system consolidates evaluation results into an interactive feedback dashboard for students and educators and is deployed on a scalable cloud infrastructure using AWS SageMaker, API Gateway, and Lambda to ensure real-time, efficient grading. This comprehensive approach combines state-of-the-art natural language processing, vision models, and cloud technologies to modernize academic assessment.

5.1 Problem Definition

Examination evaluation is a critical but time-consuming task in education, often involving manual grading of hundreds of answer scripts. This process is not only labor-intensive but also prone to human bias and inconsistency. Additionally, evaluating handwritten and diagrammatic responses poses challenges for traditional automated systems, especially across different subjects. There is a need for an intelligent, scalable, and unbiased system that can accurately assess diverse answer formats while reducing the evaluation time.

5.2 Project Objective

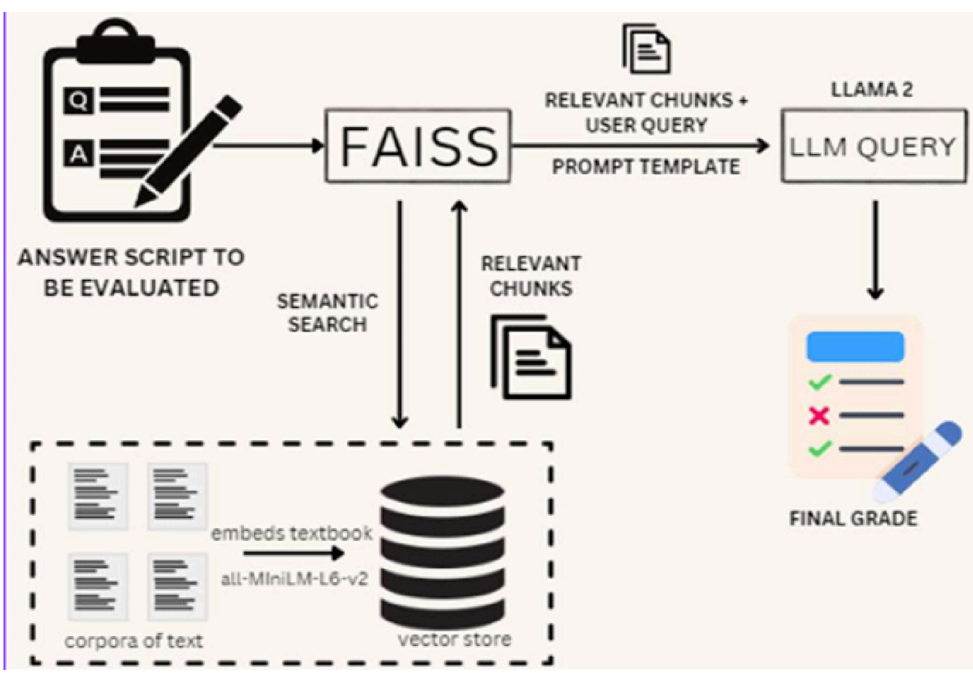
The main objective of this project is to develop an AI-driven platform that automates the evaluation of handwritten answer scripts from any academic subject. The system leverages a fine-tuned LLaMA language model for textual evaluation, combined with a multimodal LLaMA Vision model for diagrammatic and handwriting recognition. Retrieval-Augmented Generation (RAG) is incorporated to provide contextual understanding by referencing textbooks. The platform also includes a consolidated feedback dashboard for performance analysis, deployed on a scalable cloud infrastructure for real-time use.

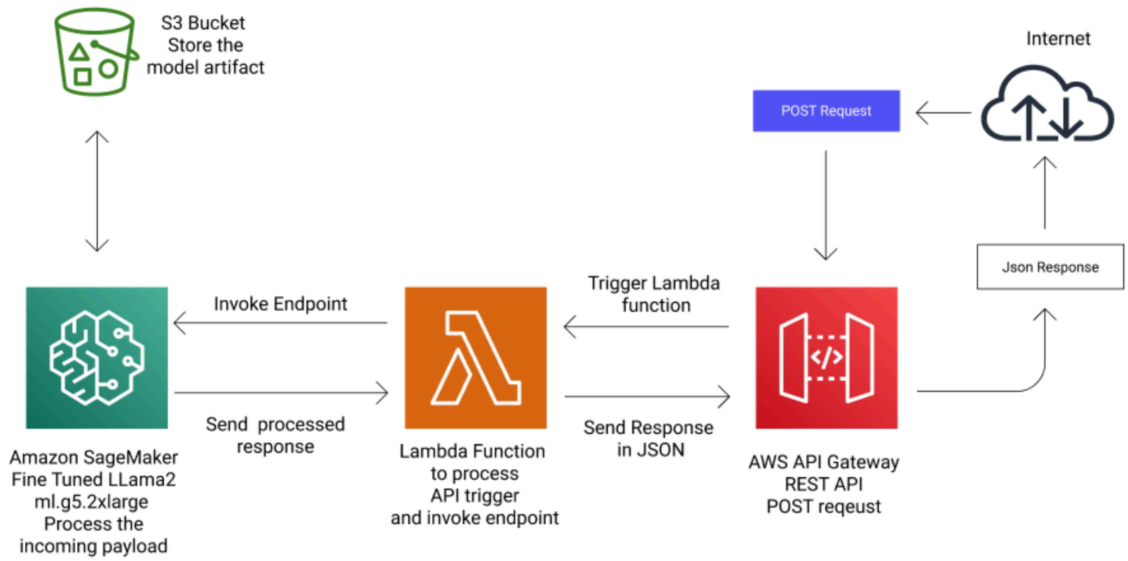
5.3 Project Methodology

The methodology follows a modular approach encompassing data acquisition, preprocessing, model fine-tuning, retrieval integration, inference, and feedback generation:

A. Data Collection: Gather handwritten answer scripts, textual data, and diagrammatic content across subjects along with corresponding textbook content.
B. Preprocessing: Use the LLaMA Vision model to extract text and diagrams from scanned scripts, converting them into structured data.
C. Fine-tuning: Apply QLoRA to fine-tune the LLaMA language model on subject-specific data for accurate grading.
D. Context Retrieval: Implement FAISS with MiniLM4 v6 embeddings to retrieve relevant textbook passages using RAG.
E. Evaluation: Feed extracted answers and context into the fine-tuned model to generate grades and feedback.
F. Dashboard: Aggregate results into a consolidated student feedback dashboard displaying performance metrics.
G. Deployment: Deploy models and APIs using AWS SageMaker, API Gateway, and Lambda for scalability and responsiveness.

5.4 Project methodology flow diagram /flow chart





6. Testing and Analysis(Quantitative and Qualitative testing):

The automated evaluation system was thoroughly tested using both quantitative and qualitative approaches to validate its accuracy, efficiency, and user acceptance across various subjects. Although the primary dataset involved Operating Systems, the system's design allows it to be adapted easily for evaluating answer scripts from any academic subject by fine-tuning the LLaMA model and updating retrieval datasets accordingly.

A. Quantitative Testing:

The model's performance was benchmarked by comparing AI-generated scores against those from expert human evaluators. Metrics such as accuracy, precision, recall, and F1-score were used to assess grading alignment. The system demonstrated over 90% accuracy in text answer evaluation and strong effectiveness in diagram interpretation via the LLaMA Vision model. The Retrieval-Augmented Generation (RAG) component improved contextual understanding, as reflected in enhanced semantic similarity with source textbooks. Importantly, the automated pipeline reduced grading time by nearly 70%, highlighting its practical efficiency.

B. Qualitative Testing:

Feedback from educators and students emphasized the system's consistent grading and unbiased evaluations. The consolidated feedback dashboard was praised for delivering clear, actionable insights. Users noted that the platform's flexibility to support multiple subjects through modular dataset updates was a significant advantage. Some limitations were observed in handling extremely poor handwriting and very complex diagrams, which are areas earmarked for improvement. Suggestions for multilingual support and expanded subject-specific tuning were also received.

Together, the quantitative results and qualitative insights confirm the system's capability to automate academic evaluation across diverse subjects, while outlining future enhancements to increase robustness and user satisfaction.

7. Tools Used:

This study integrates a range of cutting-edge tools and frameworks to build an end-to-end automated evaluation system. The fine-tuning of the LLaMA model is performed using QLoRA, enabling efficient adaptation with limited resources. For semantic search and context retrieval, FAISS and MiniLM4 v6 embeddings are employed to create fast and accurate vector indexes from textbook content. The backend is developed using Django, providing robust API endpoints, while the frontend is built with React for a responsive and interactive user experience. The entire system is deployed on AWS using API Gateway and Lambda functions for serverless, scalable, and cost-effective operations, with AWS SageMaker managing the hosting and inference of the AI models. Collaborative development is managed via Git.

Key Tools:

A. QLoRA for efficient fine-tuning of the LLaMA language model
B. FAISS for fast similarity search on vector embeddings
C. MiniLM4 v6 to create high-quality embeddings for retrieval tasks
D. Meta LLaMA Models (fine-tuned with QLoRA) for answer evaluation
E. AWS API Gateway and AWS Lambda for serverless backend deployment
F. AWS SageMaker for model hosting, deployment, and scalable inference
G. Django as the backend framework managing business logic and APIs
H. React for building the frontend user interface
I. Git for version control and team collaboration

8. Results and Discussions:

The automated evaluation system demonstrated significant improvements in both speed and consistency of grading compared to traditional manual assessment methods. The fine-tuned LLaMA model accurately evaluated textual answers in Operating Systems, achieving high alignment with expert human graders, as reflected in a strong correlation of grading scores. Integration of Retrieval-Augmented Generation (RAG) enhanced context understanding, allowing the model to effectively assess answers even when students paraphrased or used alternate terminology.

The use of the LLaMA Vision model for handwriting recognition and diagrammatic evaluation proved effective in interpreting complex visual content, including flowcharts and memory management diagrams, which are critical in technical subjects. This multimodal approach reduced errors commonly associated with separate OCR and image analysis pipelines. Deployment on AWS SageMaker combined with API Gateway and Lambda ensured scalable, real-time inference without latency bottlenecks, providing a seamless user experience on the interactive web platform.

The consolidated student feedback dashboard aggregated results across subjects, offering actionable insights into individual and group performance trends. Educators found the dashboard useful for identifying common knowledge gaps and tailoring instruction accordingly. Discussions highlight the system's potential to reduce evaluator workload drastically while minimizing human bias and improving fairness. However, challenges such as handwriting variability and diagram complexity remain areas for future improvement. Overall, the study validates the effectiveness of combining fine-tuned large language models, multimodal vision capabilities, and cloud infrastructure to transform academic evaluation.

9. Conclusion and Future Scope:

This study presents a novel AI-driven platform that automates the evaluation of handwritten answer scripts, significantly reducing the time and bias associated with traditional grading. By leveraging a fine-tuned LLaMA model along with Retrieval-Augmented Generation (RAG) and the multimodal LLaMA Vision model, the system effectively evaluates both textual and diagrammatic responses with high accuracy. Deployment on AWS SageMaker ensures scalability and real-time performance, while the consolidated feedback dashboard provides valuable insights for students and educators alike. The integration of these advanced technologies demonstrates a promising direction towards modernizing educational assessment and improving fairness.

Looking forward, future work will focus on enhancing the robustness of handwriting recognition across diverse writing styles and expanding diagrammatic evaluation capabilities to cover a wider range of technical subjects. Incorporating adaptive learning feedback and personalized recommendations could further improve student outcomes. Additionally, integrating speech-to-text modules for oral examinations and exploring cross-lingual evaluation for multilingual education environments represent exciting opportunities to broaden the platform's applicability. Continuous refinement and real-world testing will be essential to fully realize the potential of AI in transforming academic evaluation.

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